



## SAFETY DATA SHEET

### KAY-5 SANITIZER/CLEANER

#### Section: 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : KAY-5 SANITIZER/CLEANER

Other means of identification : Not applicable.

Recommended use : Sanitizer

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.32 %

Company : Ecolab Sdn. Bhd.  
Suite 12-01, 12-02 & 12-03A  
Level 12, The Pinnacle  
Persiaran Lagoon  
Bandar Sunway  
46150 Petaling Jaya  
Selangor Darul Ehsan

+603 7628 5200

Emergency telephone number : 1-800-80-1110

Issuing date : 28.05.2019

#### Section: 2. HAZARDS IDENTIFICATION

##### GHS Classification

##### Product AS SOLD

Acute toxicity (Oral) : Category 4

Acute toxicity (Dermal) : Category 4

Serious eye damage/eye irritation : Category 2B

Skin sensitization : Category 1

##### Product AT USE DILUTION

Not a hazardous substance or mixture.

##### GHS Label element

##### Product AS SOLD

Hazard pictograms :



Signal Word : Warning

Hazard Statements : Harmful if swallowed or in contact with skin  
May cause an allergic skin reaction.  
Causes eye irritation.

Precautionary Statements : **Prevention:**  
Avoid breathing dust/ fume/ gas/ mist/ vapours/ spray. Wash skin thoroughly after handling. Do not eat, drink or smoke when using this

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product. Contaminated work clothing should not be allowed out of the workplace.

**Response:**

IF ON SKIN: Wash with plenty of soap and water. Call a POISON CENTER or doctor/ physician if you feel unwell.

If skin irritation or rash occurs: Get medical advice/ attention. If eye irritation persists: Get medical advice/ attention. Wash contaminated clothing before reuse.

**Disposal:**

Dispose of contents/ container to an approved waste disposal plant.

**Product AS SOLD**

**Other hazards** : Mixing this product with acid or ammonia releases chlorine gas.

### Section: 3. COMPOSITION/INFORMATION ON INGREDIENTS

**Product AS SOLD**

Pure substance/mixture : Mixture

| Chemical Name                              | CAS-No.    | Concentration: (%) |
|--------------------------------------------|------------|--------------------|
| Sodium Carbonate                           | 497-19-8   | 10 - 30            |
| Sodium dichloro-s-triazinetriene dihydrate | 51580-86-0 | 5 - 10             |

**Product AT USE DILUTION**

No hazardous ingredients

### Section: 4. FIRST AID MEASURES

**Product AS SOLD**

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Wash clothing before reuse. Thoroughly clean shoes before reuse. Get medical attention.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Treat symptomatically. Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

**Product AT USE DILUTION**

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Rinse with plenty of water.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

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#### Section: 5. FIREFIGHTING MEASURES

##### Product AS SOLD

- Suitable extinguishing media : Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
- Unsuitable extinguishing media : None known.
- Specific hazards during firefighting : Not flammable or combustible.
- Hazardous combustion products : Decomposition products may include the following materials:  
Carbon oxides  
Sulphur oxides
- Special protective equipment for firefighters : Use personal protective equipment.
- Specific extinguishing methods : Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations. In the event of fire and/or explosion do not breathe fumes.

#### Section: 6. ACCIDENTAL RELEASE MEASURES

##### Product AS SOLD

- Personal precautions, protective equipment and emergency procedures : Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : Do not allow contact with soil, surface or ground water.
- Methods and materials for containment and cleaning up : Sweep up and shovel into suitable containers for disposal.

##### Product AT USE DILUTION

- Personal precautions, protective equipment and emergency procedures : Refer to protective measures listed in sections 7 and 8.
- Environmental precautions : No special environmental precautions required.
- Methods and materials for containment and cleaning up : Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). Flush away traces with water. For large spills, dike spilled material or otherwise contain material to ensure runoff does not reach a waterway.

#### Section: 7. HANDLING AND STORAGE

##### Product AS SOLD

- Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Do not breathe dust/fume/gas/mist/vapours/spray. Wash hands thoroughly after handling. Do not get in eyes, on skin, or on clothing. Mixing this product with acid or ammonia releases chlorine gas. In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

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Conditions for safe storage : Keep in a cool, well-ventilated place. Keep away from reducing agents. Keep away from combustible material. Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

Storage temperature : 0 °C to 40 °C

#### Product AT USE DILUTION

Advice on safe handling : In case of mechanical malfunction, or if in contact with unknown dilution of product, wear full Personal Protective Equipment (PPE).

Conditions for safe storage : Keep out of reach of children. Keep container tightly closed. Store in suitable labeled containers.

### Section: 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

#### Product AS SOLD

##### Components with workplace control parameters

| Components | CAS-No.   | Form of exposure | Permissible concentration        | Basis  |
|------------|-----------|------------------|----------------------------------|--------|
| chlorine   | 7782-50-5 | TWA              | 0.5 ppm<br>1.5 mg/m <sup>3</sup> | MY OEL |

Contains no substances with occupational exposure limit values.

Engineering measures : Effective exhaust ventilation system. Maintain air concentrations below occupational exposure standards.

#### Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Remove and wash contaminated clothing before re-use. Wash face, hands and any exposed skin thoroughly after handling.

#### Product AT USE DILUTION

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

#### Personal protective equipment

Eye protection : No special protective equipment required.

Hand protection : No special protective equipment required.

Skin protection : No special protective equipment required.

Respiratory protection : No personal respiratory protective equipment normally required.

### Section: 9. PHYSICAL AND CHEMICAL PROPERTIES

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|                                         | Product AS SOLD                                 | Product AT USE DILUTION |
|-----------------------------------------|-------------------------------------------------|-------------------------|
| Appearance                              | : powder                                        | liquid                  |
| Colour                                  | : opaque, white                                 | colourless              |
| Odour                                   | : slight                                        | Chlorine                |
| pH                                      | : 10.0 - 10.4, (1 %)                            | 10.5                    |
| Flash point                             | : Not applicable., Does not sustain combustion. |                         |
| Odour Threshold                         | : no data available                             |                         |
| Melting point/freezing point            | : no data available                             |                         |
| Initial boiling point and boiling range | : no data available                             |                         |
| Evaporation rate                        | : no data available                             |                         |
| Flammability (solid, gas)               | : no data available                             |                         |
| Upper explosion limit                   | : no data available                             |                         |
| Lower explosion limit                   | : no data available                             |                         |
| Vapour pressure                         | : no data available                             |                         |
| Relative vapour density                 | : no data available                             |                         |
| Relative density                        | : 1.0 - 1.5                                     |                         |
| Water solubility                        | : soluble                                       |                         |
| Solubility in other solvents            | : no data available                             |                         |
| Partition coefficient: n-octanol/water  | : no data available                             |                         |
| Auto-ignition temperature               | : no data available                             |                         |
| Thermal decomposition                   | : no data available                             |                         |
| Viscosity, kinematic                    | : no data available                             |                         |
| Explosive properties                    | : no data available                             |                         |
| Oxidizing properties                    | : Yes                                           |                         |
| Molecular weight                        | : no data available                             |                         |
| VOC                                     | : no data available                             |                         |

### Section: 10. STABILITY AND REACTIVITY

#### Product AS SOLD

|                                    |                                                                                |
|------------------------------------|--------------------------------------------------------------------------------|
| Reactivity                         | : No dangerous reaction known under conditions of normal use.                  |
| Chemical stability                 | : Stable under normal conditions.                                              |
| Possibility of hazardous reactions | : Mixing this product with acid or ammonia releases chlorine gas.              |
| Conditions to avoid                | : None known.                                                                  |
| Incompatible materials             | : Acids                                                                        |
| Hazardous decomposition products   | : Decomposition products may include the following materials:<br>Carbon oxides |

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Sulphur oxides

#### Section: 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Inhalation, Eye contact, Skin contact

##### Potential Health Effects

###### Product AS SOLD

Eyes : Causes eye irritation.

Skin : Harmful in contact with skin. May cause allergic skin reaction.

Ingestion : Harmful if swallowed.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

###### Product AT USE DILUTION

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

##### Experience with human exposure

###### Product AS SOLD

Eye contact : Redness, Irritation

Skin contact : Redness, Irritation, Allergic reactions

Ingestion : No information available.

Inhalation : No symptoms known or expected.

###### Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

##### Toxicity

###### Product AS SOLD

###### Product

Acute oral toxicity : LD50 rat: > 300 - 2,000 mg/kg  
Test substance: Product

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|                                   |                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------|
| Acute inhalation toxicity         | : 4 h LC50 rat: > 5.2 mg/l<br>Test substance: Product<br>Test atmosphere: dust/mist |
| Acute dermal toxicity             | : LD50 rabbit: > 1,000 - 2,000 mg/kg<br>Test substance: Product                     |
| Skin corrosion/irritation         | : No skin irritation<br>Test substance: Product                                     |
| Serious eye damage/eye irritation | : Mild eye irritation<br>Test substance: Product                                    |
| Respiratory or skin sensitization | : May cause sensitisation by skin contact.<br>Test substance: Product               |
| Carcinogenicity                   | : no data available                                                                 |
| Reproductive effects              | : no data available                                                                 |
| Germ cell mutagenicity            | : no data available                                                                 |
| Teratogenicity                    | : no data available                                                                 |
| STOT - single exposure            | : no data available                                                                 |
| STOT - repeated exposure          | : no data available                                                                 |
| Aspiration toxicity               | : no data available                                                                 |

### Section: 12. ECOLOGICAL INFORMATION

#### Product AS SOLD

##### Ecotoxicity

Environmental Effects : Toxic to aquatic life.

##### Product

Toxicity to fish : no data available

Toxicity to daphnia and other aquatic invertebrates : no data available

Toxicity to algae : no data available

##### Components

Toxicity to fish : Sodium Carbonate  
96 h LC50 *Lepomis macrochirus* (Bluegill sunfish): 300 mg/l

##### Components

Toxicity to daphnia and other aquatic invertebrates : Sodium Carbonate  
48 h EC50 *Ceriodaphnia* (water flea): 213.5 mg/l

Sodium dichloro-s-triazinetriene dihydrate  
48 h EC50 *Daphnia*: 0.196 mg/l

##### Persistence and degradability

#### Product AS SOLD

Readily biodegradable.

#### Product AT USE DILUTION

Readily biodegradable.

##### Bioaccumulative potential

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no data available

#### Mobility in soil

no data available

#### Other adverse effects

no data available

### Section: 13. DISPOSAL CONSIDERATIONS

#### Product AS SOLD

Disposal methods : Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

The product should not be allowed to enter drains, water courses or the soil.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

#### Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

### Section: 14. TRANSPORT INFORMATION

#### Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

#### Land transport

Not dangerous goods

#### Sea transport (IMDG/IMO)

Not dangerous goods

### Section: 15. REGULATORY INFORMATION

#### National Regulations

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulation 2013  
Occupational Safety and Health Act 1994  
Environmental Quality Act 1974

The components of this product are reported in the following inventories:

#### United States TSCA Inventory :

All substances listed as active on the TSCA inventory



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**Canadian Domestic Substances List (DSL) :**

All components of this product are on the Canadian DSL.

**Australia. Industrial Chemical (Notification and Assessment) Act :**

On the inventory, or in compliance with the inventory

**New Zealand. Inventory of Chemicals (NZIoC), as published by ERMA New Zealand :**

not determined

**Japan. ENCS - Existing and New Chemical Substances Inventory :**

On the inventory, or in compliance with the inventory

**Korea. Korean Existing Chemicals Inventory (KECI) :**

On the inventory, or in compliance with the inventory

**Philippines Inventory of Chemicals and Chemical Substances (PICCS) :**

On the inventory, or in compliance with the inventory

**China Inventory of Existing Chemical Substances :**

On the inventory, or in compliance with the inventory

**Taiwan Chemical Substance Inventory :**

On the inventory, or in compliance with the inventory

### Section: 16. OTHER INFORMATION

|              |                      |
|--------------|----------------------|
| Issuing date | : 28.05.2019         |
| Version      | : 1.4                |
| Prepared by  | : Regulatory Affairs |

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the SDS.

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.